Fiscal Year:	FY 2024	**Task Last Updated:**	FY 08/29/2023
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	BRIC: Exploring Spaceflight-Linked Changes in Plant Defense Capabilities		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential (2) Bioregenerative Life Support		
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Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Flight	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	10/26/2018	**End Date:**	10/25/2024
No. of Post Docs:	3	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA KSC
Contact Monitor:	Massa, Gioia	**Contact Phone:**	321-861-2938
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 10/25/2024 per NSSC information (Ed., 10/16/23) NOTE: End date changed to 10/25/2023 per NSSC information (Ed., 11/3/22) NOTE: End date changed to 10/25/2022 per NSSC information (Ed., 1/4/22) NOTE: End date changed to 10/25/2021 per G. Massa/KSC; forthcoming in NSSC (Ed., 9/24/20)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Swanson, Sarah Ph.D. (University of Wisconsin, Madison)		
Grant/Contract No.:	80NSSC19K0126		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Analysis of the plant biology transcriptomics and proteomics data in the GeneLab data system reveals that a common feature of spaceflight is alteration in plant defense-related transcripts and proteins in space. Anecdotally, plants have also been seen to suffer from pathogen attack during spaceflight missions. Taken together, these observations suggest a hypothesis that plant defense responses may be compromised by spaceflight. Such a disruption of defense would have important implications for the management of plants grown as part of an astronaut life support system. This is especially relevant given the large microbial community on board the International Space Station (ISS) and the possibility that spaceflight triggers enhanced microbial pathogenicity. However, the critical unanswered question is whether the altered defense gene expression seen in spaceflight truly plays out as altered susceptibility to pathogens. Use of the Biological Research in Canisters-Light Emitting Diodes (BRIC-LED) hardware growing Arabidopsis seedlings allows direct testing of this idea. Specifically, the project will: (1) Capitalize on the unique capabilities of the BRIC-LED hardware to analyze the transcriptome and cell wall structure responses of Arabidopsis seedlings grown in spaceflight. Plants will be grown in the hardware and then challenged with the bacterial elicitor peptide flg22. Comparison of subsequent on-orbit responses in wild-type and mutants in the defense regulators AtRBOHD and CML24 will then be made. (2) Use comparisons to ground-based control experiments and clinostat and hypoxia treatment to ask whether such changes are likely linked to the microgravity or hypoxic elements of spaceflight and whether these changes may affect the ability of the plant to defend itself against pathogen challenge. The BRIC-LED/Petri Dish Fixation Unit (PDFU) hardware is well-suited to this experimental approach. Thus, the two-stage PDFU hardware will allow both stimulation (injection of flg22 elicitor onto the seedlings) and subsequent fixation of samples (with RNAlater). As the experiment would use flg22 peptide to elicit defense response, potential problems with flying live pathogen will be circumvented. The experimental design will capitalize on the minimal crew time needed for a BRIC-based experiment and the unique capabilities of the BRIC hardware to treat plants in situ on orbit. The results from this experiment will allow direct testing of the critical question of whether the ability of plants to mount a defense response is compromised during spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is addressing how pathogen sensing and response operates in plants, targeting the Ca2+-dependent and reactive oxygen signaling machinery. The results from this work are not only providing insight into how pathogen signals impact plant growth and defense responses in space but also are contributing to our understanding of similar responses on Earth. Pathogen attack is a major limit on crop productivity with estimates of global losses in the range of 20% across multiple major crops such as wheat, maize, rice, soybean, and potato. The pathogen sensing and response systems triggered by the molecular signals of bacterial attack in space are also those employed by plants challenged by pathogens on Earth. Thus, this work aimed at analysis of how reactive oxygen species and Ca2+ signaling is involved in the defense response systems induced during spaceflight should also help our understanding for these same systems in this general plant defense response system on Earth. These results will provide information to highlight novel molecular targets for potential manipulation to tailor plant responses to the wide spectrum of pathogens that attack plants.
Task Progress:	The flight component of this project is currently waiting on the availability of the Biological Research in Canisters-Light Emitting Diodes (BRIC-LED) hardware and flight opportunities. Fresh seed stocks are being maintained for all the genotypes for the experiment. Meanwhile an analysis of the metadata associated with all the plant spaceflight data deposited at NASA GeneLab has been completed. Details including hardware, genotype, and experimental design were curated into a data matrix that was then analyzed for similarities between experiments. The similarity score matrix was then used to define highly similar or dissimilar experiments for comparison of their transcriptomic signatures. This analysis showed that assay type (microarray versus RNA-seq) and hardware for the experiment (especially its lighting) were factors that had significant effects on transcriptional patterns that can obscure the spaceflight versus ground control comparisons. Once these variables were accounted for, spaceflight effects on patterns of gene expression related to defense, oxidative stress, hypoxia, and cell wall modifications were detected. Spaceflight is thought to impose hypoxic conditions on biology due to the loss of convection in the microgravity environment. Bioinformatics was performed on plant gene expression datasets to pinpoint potential hypoxia-related response genes that might be altered by spaceflight. This analysis highlighted the vacuolar Ca2+ pump CAX2 and the ER-localized Ca2+-pumping ATPase ACA1 as strong candidates. Analysis of the low oxygen and flooding response of mutants in both genes found that knockout of CAX2 in particular leads to resistance to both flooding and hypoxic challenge. Further, analysis of patterns of gene expression indicates that many of the hypoxia-triggered genes seen in wild type plants are unaffected by this stress in the CAX2 mutants, whereas a subset of these genes is constitutively upregulated in the CAX2 lines even without hypoxic challenge. These upregulated genes are related to the processing of lactic acid, a key byproduct of metabolism under low oxygen conditions. These results have led to a model whereby knocking out CAX2 leads to preadaptation to low oxygen stress and so to increased tolerance to hypoxic environments. The CAX2 mutants should therefore show resistance to growing under spaceflight conditions.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
Articles in Peer-reviewed Journals	Barker R, Kruse CPS, Johnson C, Saravia-Butler A, Homer H, Chang HS, Trane RM, Kinscherf N, Villacampa NA, Manzano A, Herranz R, Davin LB, Lewis NG, Perera I, Wolverton C, Gupta P, Jaiswa P, Reinsch SS, Wyatt S, Gilroy S. "Meta-analysis of the spaceflight and microgravity response of the Arabidopsis plant transcriptome." NPJ Microgravity. 2023 Mar 20;9(1):21. https://doi: 10.1038/s41526-023-00247-6 ; PubMed PMID: 36941263; PubMed Central PMCID: PMC10027818 , Mar-2023
Articles in Peer-reviewed Journals	Bakshi A, Swanson SJ, Gilroy S. "A touchy subject: Ca2+ signaling during leaf movements in Mimosa." Cell Calcium. 2023 Mar;110:102695. https://doi.org/10.1016/j.ceca.2023.102695 , Mar-2023